

House price prediction using machine learning

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ABSTRACT: In addressing the formidable challenge of predicting housing prices, this project introduces an advanced machine learning-based model designed to significantly enhance accuracy. The complexity of diverse location data poses a significant hurdle, necessitating a comprehensive approach. Our suggested methodology creates a framework that aims for more accurate predictions by combining advanced machine learning algorithms with sophisticated data pre-processing techniques. Our model's performance is demonstrated through rigorous examination utilizing real-time housing price data, demonstrating significant gains over current techniques. By offering a reliable and effective method, this research makes a substantial contribution to the field of housing price prediction. The potential impact of our model extends to facilitating better decision-making in real estate transactions, where accurate pricing predictions play a pivotal role. By addressing the intricate variability in data associated with diverse location points, our model stands as a noteworthy advancement in the quest for more reliable and insightful tools for real estate professionals and stakeholder.

Keywords: linear regression, House price prediction, Machine Learning, Random Forest

1 INTRODUCTION

Due to the notable expansion of the real estate industry in recent years, stakeholders including potential buyers, sellers, and investors are now highly concerned about housing prices. Precise house price forecasting has grown in significance because it assists buyers in making educated judgments, sellers in determining the value of their homes, and investors in identifying profitable prospects. Real estate prediction has seen a radical change with the advent of machine learning algorithms. Offering a cutting-edge strategy that enables more precise and trustworthy forecasts, whereas prior approaches depended on statistical analysis and expert knowledge. With a variety of techniques, machine learning models may find intricate relationships and patterns in large datasets. These models can produce predictions for fresh data inputs that are more accurate by utilizing this feature. A strong home price forecasting model based on machine learning technique is shown in this research. Our method includes a wide range of data, such as demographics of the neighborhood, economic indicators, and property attributes, all of which boost the model's ability to predict house values. To thoroughly train the model, A large dataset of property values and other attributes is employed, gathered from many sources including public records, real estate websites, and databases. Our primary objective is to showcase the accuracy with which machine learning algorithms can approximate the prices of homes. The objective of this study is to enhance comprehension and utilization of cutting-edge technology in the real estate sector by highlighting the revolutionary influence that machine learning may have on the precision and dependability of property value forecasts.

2 LITERATURE SURVEY

Quang Truong *et al.* Changes in the price of homes are often gauged using the Home Price Index (HPI). Individual property prices cannot be predicted just by the HPI since there is a high correlation between house prices and other characteristics like location, area, population, etc. Many publications effectively anticipate house values using traditional machine learning approaches; however, they seldom discuss the performance of particular models and neglect less popular and complex models. As a result, this study will examine the distinctions between multiple advanced models to investigate the diverse impacts of variables on prediction techniques using both conventional and sophisticated machine learning techniques. Additionally, a full validation of several model implementation strategies on regression will be provided in this research, along with an optimistic outcome for housing price prediction.

Aman Chaurasia and so on. The complexity of high data variation with changes in geographical points makes house price prediction a challenging task. Our approach for estimating house prices in this study is based on machine learning. Which we believe can improve forecast accuracy. The suggested model concurrently makes use of machine learning algorithms and data pre-processing methods. Using data on current home prices, the suggested model's effectiveness is assessed, and the findings demonstrate a notable improvement over the methods used previously.

Nor Hamizah Zulkifley et al. These days, the real estate sector makes extensive use of data mining. Since data mining can extract relevant information from raw data, it is very useful in estimating property prices and significant housing qualities, among many other things. Research has shown that homeowners and the real estate sector often experience anxiety when property values fluctuate. The best successful models for forecasting both the relevant features and property prices are identified through a survey of the literature. The investigation's findings verified that the XGBoost, Support Vector Regression, and Artificial Neural Network models were the most successful when compared to other models. Our findings also suggest that structural and locational factors are important in influencing property prices. This study. The identification of the key variables impacting house prices and the best machine learning model to employ in such research would be of particular interest to housing developers and researchers.

3 METHODOLOGY

A framework uses machine learning techniques to predict price ratings by pre-processing text input, extracting relevant features, and modeling connections between reviews and ratings. The system provides a reliable and effective means of housing price predictions based on methods, textual content analysis and user sentiments.

3.1 *Proposed system*

Conversely, the goal of the suggested house price prediction system, which consists of really basic housing market models, is to provide consumers with a dependable and user-friendly tool for forecasting property prices. Users enter vital information about a home through an intuitive interface. Additional insights into market trends and environmental aspects are provided by the system's seamless integration with external data sources. Sturdy data preparation methods take care of missing values, modify categorical variables, and validate input. The core component of the system is a machine learning model, which uses regression methods to generate precise forecasts after being trained on historical housing data. Strict evaluation measures, such as R square and mean absolute error, guarantee the model's accuracy. Its performance is further enhanced by hyper-parameter adjustment. The stage of deployment provides the to the lifespan and applicability of the system. Users, developers, and maintainers are guided by thorough documentation, and security mechanisms protect

user data. Over time, the adaptability and efficacy of the system are improved by a feedback mechanism and scheduled updates based on user input and market conditions.

Figure 1 shows the Workflow Architecture.

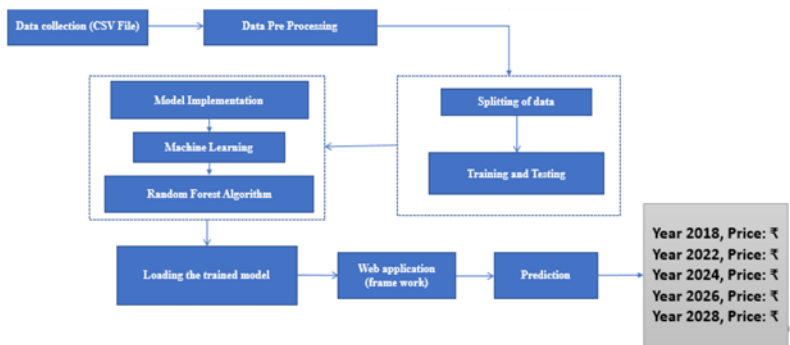


Figure 1. Proposed workflow architecture.

3.1 Information gathering

Characteristics and Aspects of Data That Are Important for Forecasting Home Prices
Accurate and trustworthy forecasting models are mostly dependent on the techniques applied to gather from Kaggle and extract the dataset, as well as the features and qualities selected for the data.

3.2 Preprocessing of data

In machine learning, validation techniques are used to derive an error rate from an ML model, which is believed to be nearly identical to the data set's true error rate. Validation strategies are unnecessary when data sizes are high enough to accurately represent the population. In real-world scenarios, it is imperative to work with data samples that do not exactly reflect the population in a particular data collection. To determine if a variable is an integer or a float, duplicate the value and supply the data type. By modifying the model's hyperparameters, a data model can offer an objective assessment of a model's fit to a training data set. The model architecture's incorporation of the validation data gathering performance makes the estimate more flexible. A regularly updated model is used to assess the model validation set. Machine learning developers utilize this data to adjust the model's upper bounds. Gathering and analyzing data, as well as managing its quality, content, and organization, can become into time-consuming to-do lists. During the data identification step, we help you understand your data and its attributes. With this insight, you can choose the method to use to build your model. Because it can swiftly clean data and is optimized for big data cleaning tasks without values, the Python library is widely used for data cleaning tasks. The processes of data cleansing, analysis, and modelling take a lot of time occasionally, mistakes could occur in any of these sources. Complex reasons can occasionally be the cause of missing data. It is crucial to comprehend these various kinds of missing data from a statistical standpoint. It goes over the many kinds of missing data, how to find and close gaps, how to do some simple accusations, and a thorough statistical analysis of missing data.

3.3 Information projection

Machine learning and applied statistics both heavily rely on knowledge of information. In actuality, statistics' main fields of study are quantitative analysis and data evaluations. A useful method for gaining qualitative insights is graphing data. It can help you spot outliers, skewed data, trends, and more while researching a data set. Data visuals can be useful in

underscoring and illuminating important connections in captivating plots and diagrams, as opposed to those that are significant or connected to a narrow body of knowledge. If you're interested in learning more about exploratory data analysis and data representation, I highly suggest reading the books I listed at the end. Without being displayed in a visual format, such graphs and charts, data can occasionally lack meaning. In applied statistics and applied machine learning, the ability to rapidly visualize data and other patterns is crucial. You will learn about the many kinds of layers that you need to be aware of when using Python to visualize data and how to use them to have a deeper understanding of your own data. How to use line and bar graphs to interpret time series data with categorical measurements. How to use boxplots and histograms to summarize data distributions.

3.4 *Random forest technique*

In the realm of real estate, harnessing the power of advanced machine learning techniques is indispensable for predicting housing prices accurately. One such formidable approach is the application of Random Forest techniques, a sophisticated ensemble learning method that excels in handling diverse datasets. To enhance the predictive capabilities of our model, a meticulous preprocessing of textual input is undertaken. This involves the extraction of pertinent features from housing-related text data, leveraging the nuances embedded within the textual information. Notably, sentiments conveyed in property descriptions and discernible patterns observed in housing narratives become pivotal indicators in informing the predictive prowess of the model. The Random Forest model is constructed, capable of navigating the complexity inherent in housing market dynamics. The ensemble of decision trees collaboratively captures the intricacies of the relationships between features, resulting in a predictive model that demonstrates a high degree of accuracy in forecasting property values. In essence, this holistic approach integrates sentiment analysis and feature extraction from textual data, culminating in a Random Forest model that stands as a formidable tool in the realm of housing price prediction, adept at deciphering the intricate interplay of factors that influence property values.

4 RESULTS AND DISCUSSION

The efficacy of models in extracting pertinent information and interpreting sentiment is demonstrated by their ability to predict price evaluations from user reviews with accuracy. Through the examination of textual data, the model pinpoints crucial attributes that impact house rate assessments, thereby advancing a refined comprehension of client encounters. By using user feedback to forecast prediction. More accurately, this strategy helps Real estate providers make well-informed decisions. The efficacy of models in extracting pertinent information and interpreting sentiment is demonstrated by their ability to predict price evaluations from user reviews with accuracy. Through the examination of textual data, the model pin-points crucial attributes that impact house rate assessments, thereby advancing a refined comprehension of client encounters. By using user feedback to forecast prediction. More accurately, this strategy helps Real estate providers make well-informed decisions.

5 CONCLUSIONS AND FUTURE ENHANCEMENT

In conclusion, the real estate sector relies heavily on accurate house price forecasting, which makes trustworthy instruments for assessing market movements essential. Machine learning algorithms have the potential to produce accurate predictions based on past data patterns; the random forest algorithm is a prime example of this. The adaptable algorithm Random Forest offers a solid method for deciphering correlations between variables and the home price goal variable. Despite its interpretability and simplicity, the efficiency of the random forest model relies on high-quality input data. Meticulous data collection, preprocessing,

and model implementation are indispensable for success. While random forest offers advantages, continuous exploration of alternative algorithms is essential to adapt to dynamic real estate markets. Ultimately, the synergy between advanced machine learning techniques and high-quality data unlocks predictive modeling's potential in real estate.

Future enhancements for a housing price prediction model using machine learning can focus on refining and expanding the model's capabilities to provide more accurate and valuable insights. One avenue for improvement is incorporating advanced feature engineering techniques to better capture the nuanced factors influencing property prices. This may involve extracting insights from unstructured data sources such as text descriptions, images, or even social media sentiment related to the neighborhoods. Moreover, the incorporation of real-time data feeds helps improve the model's adaptability to market fluctuations and guarantee that forecasts correspond with the latest patterns. This could involve leveraging APIs to gather up-to the minute information on economic indicators, local development projects, or changes in neighborhoods demographics.

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